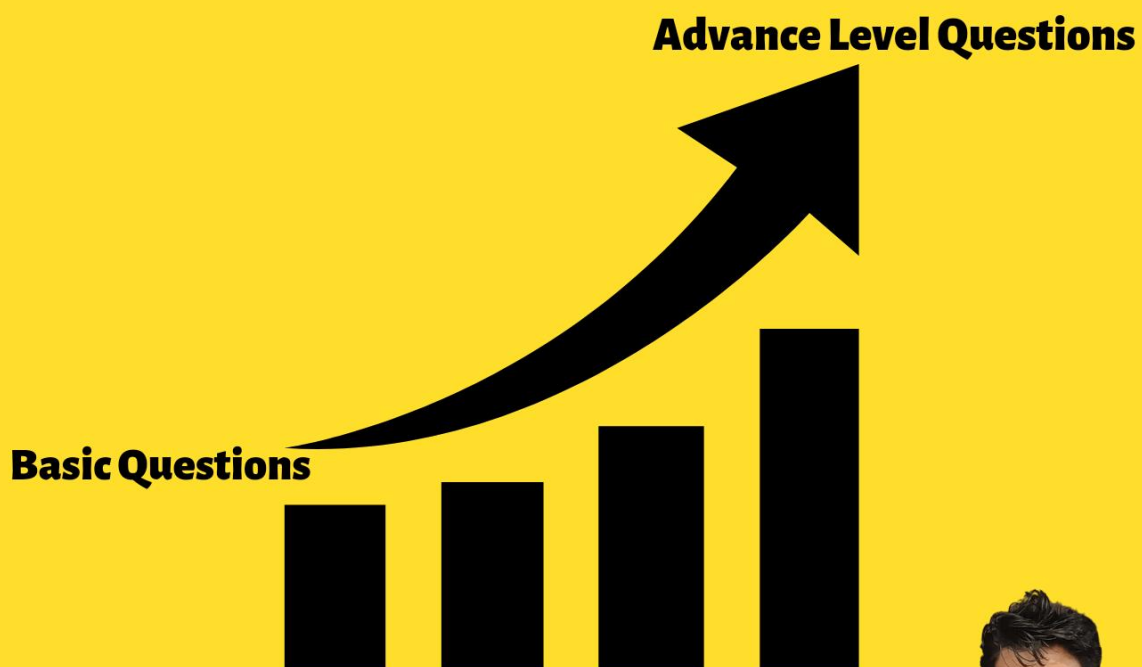


MATHS

AGE

ALL TYPE WISE QUESTIONS



Maths with Gajanand



Age

- 1] The ages of Fatima and Ahmad are in the ratio 3:8. The sum of their present ages is 44 years. The difference of their ages is:

Soln:- Basic:-

Fatima : Ahmad
3x 8x

$$3x + 8x = 44$$

$$11x = 44$$

$$x = 4$$

$$8x - 3x = 5x$$

$$= 5 \times 4$$

$$= \boxed{20 \text{ years}}$$

m-II

fatima : Ahmad

3 : 8

$$\begin{array}{r} 3 \\ + 8 \\ \hline 11 \end{array} \rightarrow 44$$

$$1 \rightarrow 4$$

$$\text{Sum} = 11 \rightarrow 11 \times 4 = \boxed{44} \text{ Ans}$$

- 2] The sum of the age of a father and his son is 100 years now. 5 years ago their age were in the ratio of 2:1. The ratio of the age of father and son after 10 years will be:

Soln:-

5 year ago →

F	S
2	1
↓	↓
60yr	30yr

$$3 \rightarrow 90 \text{ years}$$

$$1 \rightarrow 30 \text{ years}$$

$$\begin{array}{r} \downarrow +15 \\ 75 \end{array} \quad \begin{array}{r} \downarrow +15 \\ 45 \end{array}$$

$$75 : 45$$

$$\boxed{5 : 3} \text{ Ans}$$

- 3] The ratio of the age of a father to that of his son is 5:2. If the product of their age in years is 1000, then the father's age (in years) after 10 years will be:

Soln:-

$$\begin{array}{rcl}
 & F & : \quad S \\
 & 5x & \quad 2x \\
 & \swarrow & \\
 & 10x^2 = 1000 & \\
 & x = 10 & \\
 & 50 & \\
 & +10 & \\
 \text{after } \rightarrow & \boxed{60 \text{ Year}} & \text{Ans} \\
 10 \text{ year} & &
 \end{array}$$

- 4] The ratio of ages of father and son is 7:2. Five year ago the product of their ages was 150. What is the age of the father?

Soln:-

$$\begin{array}{rcl}
 & F & : \quad S \\
 & 7x & : \quad 2x \\
 & \Rightarrow (7x-5)(2x-5) = 150 & \\
 \Rightarrow & 14x^2 - 45x + 25 = 150 & \\
 \Rightarrow & 14x^2 - 45x - 125 = 0 & \\
 \text{Solve} & \boxed{x=5} & \text{so father age} = 7x = \boxed{35} \text{ Ans}
 \end{array}$$

m-II

$$\begin{array}{rcl}
 \text{Father} & & \text{son} \\
 7 & : & 2 \\
 \downarrow \times 5 & & \downarrow \times 5 \\
 35 & & 10 \\
 -5 \left(& & \right) -5 \\
 30 & & 5
 \end{array}$$

A.T.Q. $30 \times 5 = 150 \text{ years}$

Present age of Father = $\boxed{35 \text{ years}}$ Ans

[5] The brother is elder to his sister by 6 years. Seven years ago the product of their ages was 72. What is the age of the brother?

(a) 13 Year (b) 12 Year (c) 6 year (d) 19 Years

Soln:-

	Brother	Sister
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Present age $\rightarrow$ $(x+6)$ x

7 Year ago $\rightarrow$ $(x-1)$ $(x-7)$

$$(x-1)(x-7) = 72$$

by option $\boxed{x=13}$ Ans

[6] The ratio between the present ages of A and B is 3:5. If the ratio of their ages five years hence becomes 13:20, then the present age of B is:

Soln:-

A	B
(3	: 5)

$\Rightarrow$

A	B
21	: 35
+5	+5
26	40

(13 : 20) $\times 2$

B = $\boxed{35 \text{ years}}$ Ans

[7] The current ages of Sonali and Monali are in the ratio 5:3. Find five years from now, their ages will be in the ratio 10:7. Then, Monali's current age is:

Soln:-

S	M
(5	: 3)

$\Rightarrow$

S	M
15	9
+5	+5
20	14

Monali age = $\boxed{9 \text{ years}}$

Ans

8] The ratio of present ages of R and S is 11:17, before 11 years the ratio of their ages was 11:20. What is R's present age?

Soln:-

	R	S		R	S	
11 yrs before $\rightarrow$	(11	: 20)	$\times 2$	22	: 40	
			$\Rightarrow$	$\downarrow +11\text{yr}$	$\downarrow +11\text{yr}$	
Present $\rightarrow$	(11	: 17)	$\times 3$	33	: 51	R = 22 years
						<u>Ans</u>

9] The ratio of the present ages of A and B is 8:15. Eight years ago, the ratio of their ages was 6:13. What will be the ratio of ages of A and B after 8 years from now?

Soln:-

	A	B	
	(6	: 13)	$\times 2$
	$\downarrow 2$	$\downarrow 2$	2 Unit $\longrightarrow$ 8 year
Present $\rightarrow$	(8	: 15)	$\times 1$
			1 Unit $\longrightarrow$ 4 year

	A	B
Present $\rightarrow$	8	: 15
	$\downarrow \times 4$	$\downarrow \times 4$
	32 yr	60 year
	$+8$	$+8$
	40	: 68

10 : 17

Ans

10 The ratio of the age of Ram and Rahim 10 years ago was 1:3. The ratio of their age five years hence will be 2:3. Then the ratio of their present age is:

Soln:-

$$\begin{array}{lcl}
 10 \text{ Yr before } (1 : 3)_{x_1} & & 1 : 3 \\
 \text{Present -} & \Rightarrow & \begin{array}{c} | \\ \textcircled{3} \end{array} \quad \begin{array}{c} | \\ \textcircled{3} \text{ Unit} \end{array} \begin{array}{l} (10+5) \text{ Yr} \\ \downarrow \\ 15 \text{ years} \end{array} \\
 5 \text{ Yr after } (2 : 3)_{x_2} & & 4 : 6 \quad \begin{array}{l} 1 \text{ Unit} \rightarrow 5 \text{ Yr} \\ \downarrow \\ 15 \text{ year} \end{array} \\
 & & \begin{array}{cc} \times 5 \downarrow & \times 5 \downarrow \\ 20 \text{ Yr} & 30 \text{ year} \\ \downarrow -5 & \downarrow -5 \\ \text{Present year} = 15 \text{ year} & 25 \text{ year} \end{array} \\
 & & 15 : 25 \\
 & & = \boxed{3 : 5} \text{ Ans}
 \end{array}$$

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11 The ratio of the ages of A and B, four years ago, was 4:5. Eight years from now, the ratio of the ages of A and B will be 11:13. What is the sum of their present ages?

Soln:-

$$\begin{array}{lcl}
 \begin{array}{c} A \quad B \\ (4 : 5)_{x_2} \\ -4 \downarrow \\ \text{Present} \end{array} & \Rightarrow & 8 : 10 \quad (4+8=12) \\
 & & \begin{array}{c} | \\ \textcircled{3} \text{ Unit} \end{array} \rightarrow 12 \text{ year} \\
 \begin{array}{c} +8 \downarrow \\ (11 : 13)_{x_1} \end{array} & & 11 : 13 \quad 1 \text{ Unit} \rightarrow 4 \text{ Yr} \\
 & & \begin{array}{cc} \times 4 \downarrow & \times 4 \downarrow \\ 44 & 52 \\ -8 & -8 \\ \hline 36 & 44 \end{array} \\
 \text{Present} & & 36 + 44 = \boxed{80 \text{ year}} \text{ Ans}
 \end{array}$$

12 5 year ago, the ratio of the age of A to that of B was 4 : 5. Five years hence, the ratio of the age of A to that of B will be 6 : 7. If at present, C is 10 years younger than B, then what will be the ratio of the Present age of A to that of C ?

Soln:-

	A		B
	4	:	5
-5	↓		↓
	②		② Unit → 10 Year
+5	↓		↓
	6	:	7
			1 Unit → 5 year
x5	↓		↓
	30		35
	↓-5		↓-5
Present	25 yr		30 yr
	A		B

$$C = B - 10 = 30 - 10 = 20 \text{ yr}$$

$$A : C = 25 : 20$$

$$= \boxed{5 : 4} \text{ Ans}$$

13 My grandfather was 9 times older than me 16 years ago. He will be 3 times of my age 8 years from now. Eight years ago, the ratio of my age to that of my grandfather was.

Soln:-

	My		Grandfather
	1	:	9
-16	↓		↓
Present	→		→
	8	:	3
			x 8

⇒

1	:	9
		30 Unit → 24
		1 Unit → 8 yr
4	:	12
↓		↓
32 yr		96 yr
↓-16		↓-16
16	:	80

Eight year ago

$$\boxed{1 : 5} \text{ Ans}$$

- [14] The ratio of ages of the father and mother was 11:10 when their son was born. The ratio of ages of the father and ~~mat~~ mother will be 19:18 when the son will be twice his present age. What is the ratio of present ages of father and mother?

Soln:-

	Father	:	Mother
	11	:	10
x (Present)	8		8
	19	:	18

⑧ Unit = $2x$

$x = 4$

Let the present age of son be x

Ratio of Present age

$$\Rightarrow 11+4 : 10+4$$

$$\Rightarrow \boxed{15 : 14} \text{ Ans}$$

- [15] The ratio of a man's age to his father's age is 4:5, and the ratio of his age to his son's age is 6:1, four years ago these ratios were 11:14 and 11:1, respectively. The ratio of the age of the grandfather to that of the grandson 12 years from now will be:

Soln:-

Man Father Son

$$\begin{array}{ccc} 3 \times 4 & : & 5 \times 3 \\ 2 \times 6 & : & 2 \times 1 \\ \hline 12 & : & 15 : 2 \end{array}$$

$$\begin{array}{c} (-4 \text{ years ago}) \\ \hline 11 : 14 : 1 \end{array}$$

Present age =

Man	Father	Son
48	60	8

① Unit $\rightarrow$ 4 year

Grandson : Grandfather

$$(60+12) : (8+12)$$

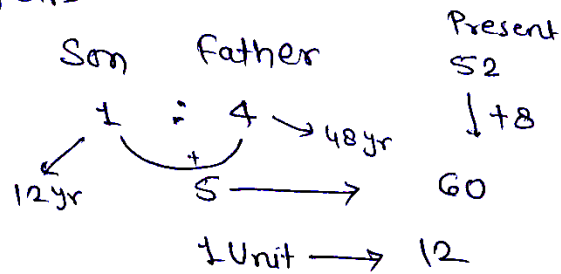
$$72 : 20$$

$$\boxed{18 : 5} \text{ Ans}$$

- 16** The sum of the present ages of a father and son is 52 years. Four years hence, the son's age will be $\frac{1}{4}$ that of the father. What will be the ratio of the ages of the son and father, 10 years from now?

Soln

after 4 years



Present age = 8 yr , 44 yr

after 10 yr = 18 : 54

= $\boxed{3 : 1}$ Ans

- 17** The present age of two persons are 36 and 50 years respectively. If after n years the ratio of their age will be 3:4, then the value of n is:

Soln: (a) 4 (b) 7 (c) 6 (d) 3

36 year 50 years

↓ +6 ↓ +6 by options

42 : 56

$\boxed{3 : 4}$ Ans

[18] A boy was asked of his age by his friend. The boy said, 'The number you get when you subtract 25 times my age from twice the square of my age will be thrice your age. If the friend's age is 14, then the age of the boy is:

Sol:- Boys Age = x $\therefore 2x^2 - 25x = 42$
 $x = \boxed{14 \text{ years}}$ Ans

[19] A says, "If you reverse my own age, the figures represent my Brother's age. He is, of course, senior to me and the difference between our ages is one-eleventh of their sum." Then A's brother's age is?

Sol:- A's age = $10y + x$
Brother = $10x + y$
diff $\rightarrow 10x + y - 10y - x = \frac{1}{11}(11x + 11y)$

$$9x - 9y = x + y$$

$$8x = 10y$$

$$\frac{x}{y} = \frac{5}{4}$$

Brother $\rightarrow 10 \times 5 + 4$
 $= \boxed{54}$ Ans